Learning Objectives:

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1. To recognize bony and soft tissue injuries to the lower extremities
2. To appreciate complications related to lower extremities
3. To understand management options in lower extremities injuries
4. To recognize bony and soft tissue injuries to the lower extremities
5. To identify urgent and emergent lower extremities conditions
6. To interpret and apply clinical signs and symptoms of lower extremities processes in the differential diagnosis.

Blood Supply to the Hip

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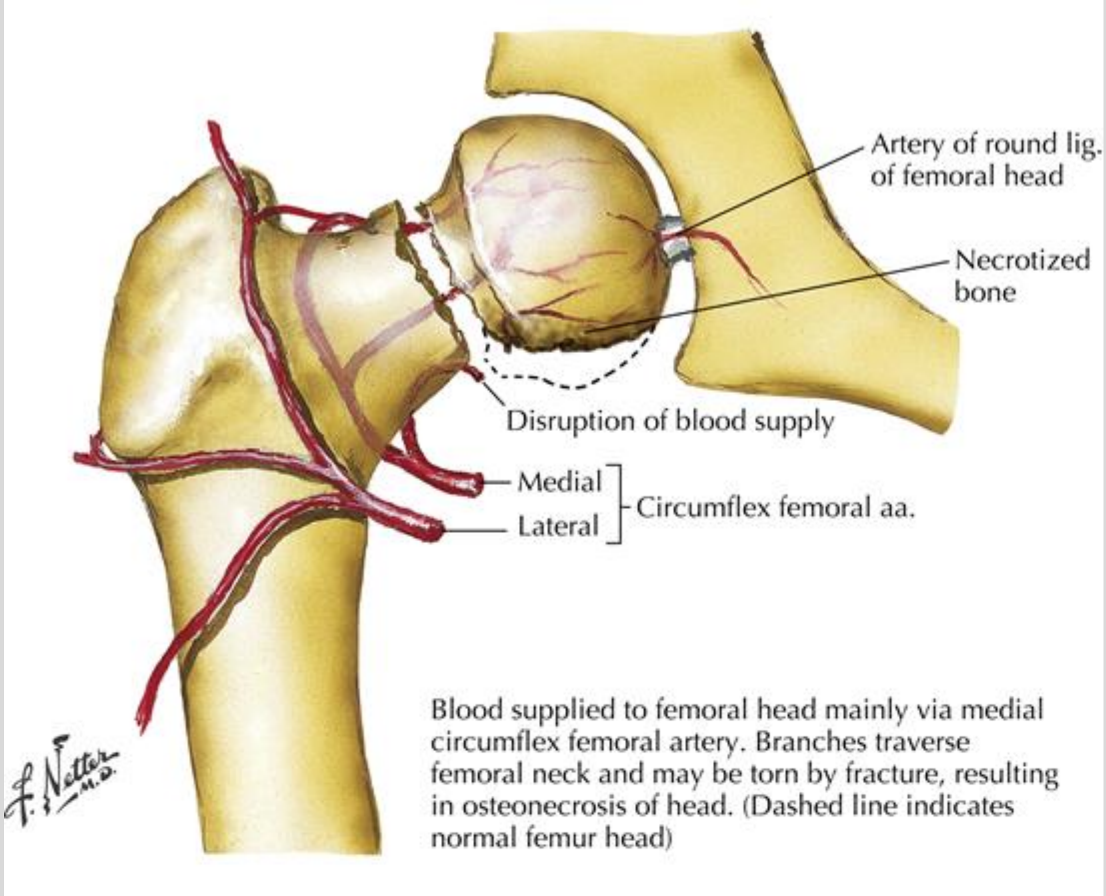

Mainly the medial circumflex femoral artery.

Intracapsular branches are the retinacular arteries.

- Cleland, Joshua A. Netter's Orthopaedic Clinical Examination. 4th Edition. 2022.

Proximal Hamstring Injuries

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a. Typically occurs during sprinting when there is a sudden stretch on the musculotendinous junction
b. Severe → Complete avulsion may occur at the ischium
c. Symptoms→ Sudden pain during exercising at the posterior thigh up to the ischium
d. May also report some numbness and tingling in the area corresponding to the sciatic nerve due to presence of hematoma that compresses the sciatic nerve
e. Treatment
    i. Strain → Supportive (RICE)
    ii. Complete Avulsion Injuries → If the hamstring is retracted more than 3 cm, early surgical intervention. If less than 3 cm of retraction, may try nonoperative treatment

Madden, Christopher. Netter's Sports Medicine. 2023

Rectus Femoris Strains- “Hip Flexor Strain”

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a. More common than complete avulsion
b. Rectus Femoris is m/c strained muscle in the quadriceps group
c. MOI: Attempt to kick a ball (football or soccer) - Higher risk when patient’s foot hits another player in the middle of the kicking motion
d. Symptoms → Acute pain in the midline of the anterior thigh
e. Pain or weakness with resisted hip flexion or knee extension
f. Treatment →
   i. Strain → Supportive (RICE) and rehab
   ii. Proximal rupture → nonoperative vs. operative

Madden, Christopher. Netter’s Sports Medicine. 2023

Femoroacetabular Impingement

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a. Results from abnormal contact between the femoral head and the acetabulum
    i. CAM (Femoral head is not round and cannot rotate smoothly inside the acetabulum)
    ii. Pincer (An acetabular abnormality whereby extra bone extends beyond the normal rim of the acetabulum, impinging upon the femoral head and the labrum)
    iii. Mixed (Combination)
b. Symptoms → Limited ROM with groin/hip pain especially with flexion and IR
c. Nonsurgical Treatments → RICE, Physical Therapy, Activity modification
d. Surgical
    i. Acetabular rim trimming
    ii. Labral repair, if needed
    iii. Labral Reconstruction, if needed
    iv. Osteochondroplasty

Madden, Christopher. Netter's Sports Medicine. 2023

Anterior Hip Dislocations

- Less common (10-15% Overall)
- History
- Physical Examination
  - Limited ROM
  - External Rotation, Flexion, Abduction
  - Careful Neurologic Evaluation

Moore's Clinically Oriented Anatomy, 9e, Ch. 7

Hip Dislocations

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1. Treatments
   a. Early reduction essential (< 6 hours)
      i. Closed reduction (without surgery, just manipulating the bone, trying to “pop” it back in)
      ii. Open reduction (surgically putting the bone in place) if irreducible or if fracture found on x-ray
   b. Repeat x-rays and Neuro exams

1. Complications
   a. Avascular necrosis
   b. Sciatic nerve injury (Posterior dislocations)
   c. Femoral artery/nerve injury (Anterior dislocations)
   d. Osteoarthritis

- Thompson, Jon C. Netter’s Concise Orthopedic Anatomy, 2nd Edition. 2010

Treatment

- Goals:
  - Protect from additional damage
  - Minimize discomfort
  - Restore hip function
  - Allow rapid mobilization
  - Avoid avascular necrosis
- Non-operative
  - Fixation almost always indicated
- Operative Management - depends on type of fracture and patient
  - Pins
  - Screws
  - Hemiarthroplasty
  - Intramedullary Nail

Femoral Shaft Fractures

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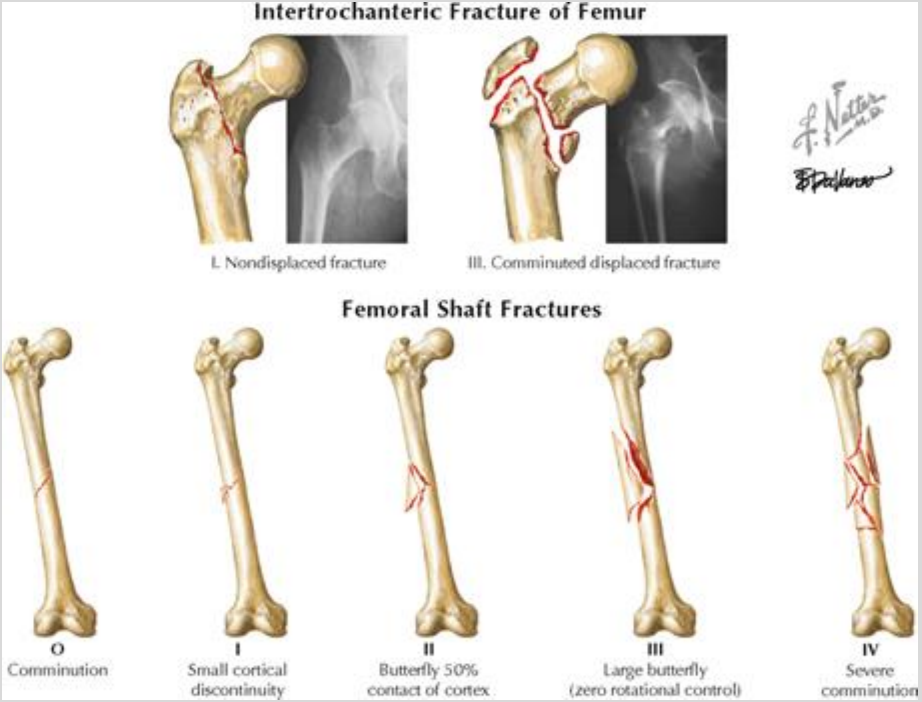
- Orthopaedic emergency
- Potential source of significant blood loss
- Monitor for Compartment Syndrome



- Thompson, Jon C. Netter's Concise Orthopedic Anatomy, 2nd Edition. 2010

Degenerative Diseases of the Hip and Knee (e.g. Osteoarthritis)

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- Pain
- Limited Range of Motion
- Functional Deficits:
    - Walking, Sitting, Standing, Dressing, Personal Hygiene, Exercise, Work

- Diagnosis: X-rays

Treatment:

- RICE, activity modification, weight loss, Physical Therapy
- NSAIDS
- Pain management? (with all of its issues)
- Joint Injections
- Surgery

Knee - Anatomy

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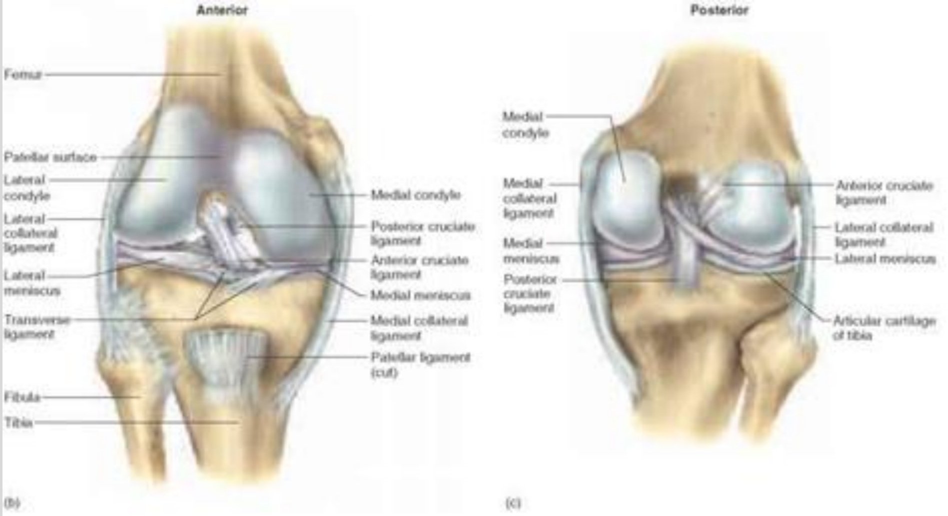
- Medial Collateral Ligament (MCL)
  - Restricts against valgus forces.
- Lateral Collateral Ligament (LCL)
  - Restricts against varus forces.
- Anterior Cruciate Ligament (ACL)
  - Restricts anterior translation & rotation of the tibia on the femur.
- Posterior Cruciate Ligament (PCL)
  - Restricts posterior translation of the tibia on the femur.
- Menisci (medial & lateral)
  - Wedge shaped fibrocartilage on the articular surfaces of the tibia
  - Shock absorption for load bearing on knee.
  - Deepen tibial surface -> aids in knee stability



Moore, K. L., Dalley, A. F.,&Agur, A. M. (2014). Lower Limb. In *Clinically Oriented Anatomy*(7th ed., p. 636-642). Lipincott Williams and Wilkins.

Anterior Cruciate Ligament (ACL) Tear

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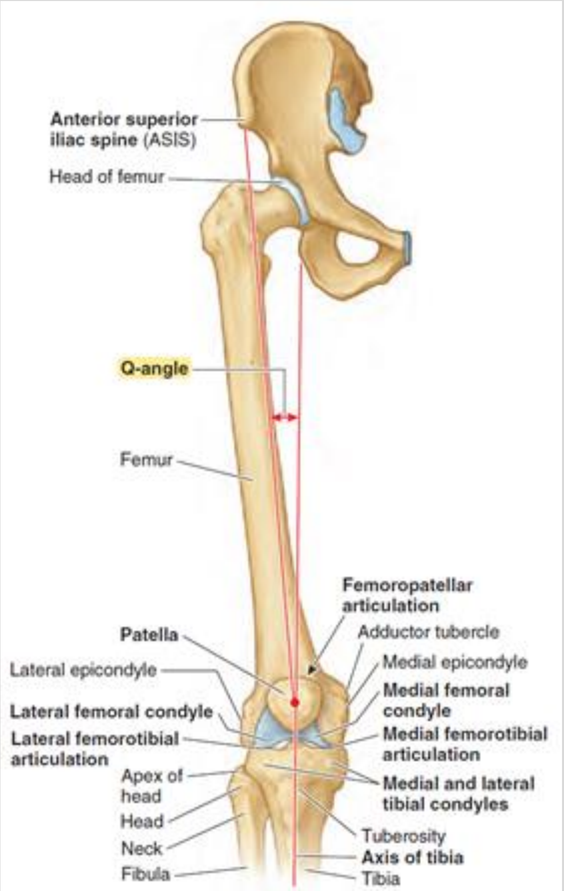
- Common, comprising 40-50% of all ligamentous knee injuries
- Female athletes 2-8x higher risk of ACL tears
  - Wider hips -> ↑Q-angle -> Greater valgus knee loading in landing
- <u>Cause-</u> Hyperextension and severe force directed anteriorly with the knee semiflexed, causing tibia to slide anteriorly against the femur. "Popping" noise is often heard when the injury first occurs.
  - Non-contact pivot/changing direction (Most common), stopping suddenly, direct blow, poor landing from a jump often cause ACL rupture.
  - Often a NONCONTACT athletic injury
- <u>Symptoms-</u>Pain, swelling, decreased ROM, discomfort while walking. Hemarthrosis (bleeding into joint space) is often found on aspiration.
- <u>Examination-</u> + Lachman test (most sensitive), + anterior drawer test, X-Ray to rule out any associated fractures, MRI for confirmation
- <u>Treatment</u>:
  - Low demand pt (e.g. elderly): PT + lifestyle mods
  - Active pt: Surgical reconstruction



Source: 1. Miller's Review of Orthopaedics: Seventh Edition. 349-352.
2. Netter's Orthopaedics. First Edition: 419-420
3. Moore, K. L., Dalley, A. F.,&Agur, A. M. (2014). Lower Limb. In *Clinically Oriented Anatomy*(7th ed., p. 663). Lipincott Williams and Wilkins.

Posterior Cruciate Ligament (PCL) Tear

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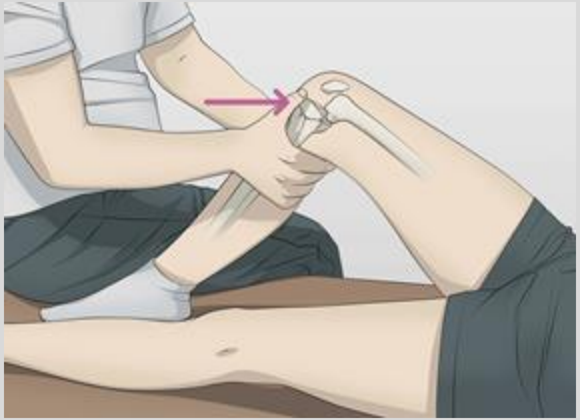
- The PCL is a strong ligament, so ruptures rarely occur.
- Causes: Contact injury (usually)
  - Fall/blow to proximal tibia w/ flexed knee (MVA dashboard injury) - tibia moves posteriorly against the femur.
  - Hyperextension injury
- Symptoms- Pain, swelling, difficulty walking, instability
- Examination- +Posterior drawer test, X-Ray to rule out concurrent bone fractures, MRI to confirm diagnosis.

Treatment- Surgery only needed when concurrent injuries (bone fractures, LCL, or MCL tears) have occurred.

Otherwise heals well with RICE (rest, ice, compression, elevation) protocol, Rehab, and bracing.



Source: 1. Moore, K. L., Dalley, A. F.,&Agur, A. M. (2014). Lower Limb. In *Clinically Oriented Anatomy*(7th ed., p. 663). Lipincott Williams and Wilkins.

https://orthoinfo.aaos.org/en/diseases--conditions/posterior-cruciate-ligament-injuries/

Knee - Bursitis, Cont.

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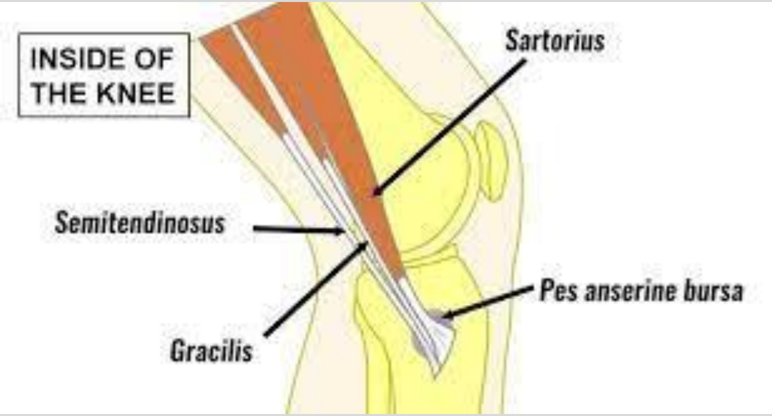
- Anserine Bursitis (Pes anserinus pain syndrome)
  - Etiology: Obesity, overuse in athletes
  - Sharp, non-radiating pain on medial side of tibia just below the knee (semitendinosus, gracilus, saritorius)
  - No pain or laxity with valgus stress test (r/o MCL)



Baker's Cyst (Popliteal Cyst)

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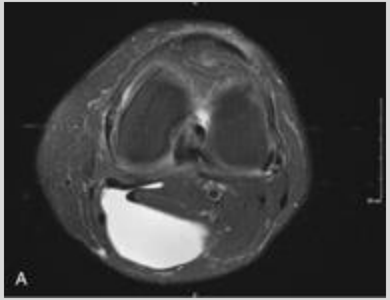
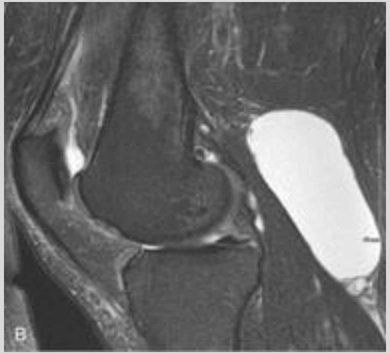
- Fluid collection in popliteal bursa behind the knee
  - Cyst forms in the bursa underlying the Gastrocnemius & Semimembranosus tendons
- This bursa communicates directly with synovial space of knee joint. Chronic joint disease of the knee (e.g. any form of arthritis) causes fluid collection in this bursa.





- <u>Common causes:</u>
  - Osteoarthritis
  - Rheumatoid arthritis
  - Joint injury (meniscal tears, ACL rupture)
  - Gout/pseudogout
- <u>Symptoms:</u>
  - Generally asymptomatic; May cause posterior knee pain, pain with prolonged standing or activity
  - Rupture causes acute severe pain that can mimic symptoms of DVT
- <u>Diagnostic Imaging:</u>
  - Ultrasound (r/o DVT); MRI

Miller's Essential Orthopaedics, 2010 Chapter 152, 657-659

DVT/VTE

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- Symptoms:
  - U/L leg pain (eg Calf pain)
  - Palpable Cord felt on back of leg (from a thrombosed vein)
  - Unilateral edema (poor venous outflow from leg)
  - Warmth, tenderness, erythema in calf or thigh
  - + Homan's Sign: increased calf pain with ankle dorsiflexion
- Diagnostic Tests:
  - Use Wells' Criteria to aid in diagnosis of DVT/PE
  - Lower extremity Compression ultrasound w/ doppler flow studies (aka Duplex Ultrasonography)
    - Initial diagnostic test of choice -> look for non-compressibility of affected vein
  - D-dimer level: fibrin breakdown product -> ↑clotting -> ↑clot breakdown -> ↑D-dimer
- Treatment:
  - Anticoagulation with unfractionated heparin or LMWH (Transition to Warfarin for LT prevention)
- Prophylaxis:
  - Early fracture stabilization & mobilization
  - Mechanical venous circulatory assist (calf pumps/SCD/IPC)
  - Pharmacologic prophylaxis (anti-coagulants)

Miller's Essential Orthopaedics, 2010 Chapter 20, 92-94

Achilles Tendon Rupture

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- Causes: Frequently seen in athletes, particularly males in 4th decade. Usually in zone about 6 cm above the calcaneal insertion (Hypovascular region of the tendon)
- Presentation: Usually occurs during push-off during running or a hard lateral movement. May hear “POP” as a sensation of getting hit in the back. Unable to push off with the ankle. Localized posterior ankle pain
- Examination: Loss of resistance to passive dorsiflexion or loss of active plantar flexion. Palpable gap. Bruising. Thompson Test.
- Treatment: Non-surgical→ Begins with immobilization in plantarflexion, with progressive dorsiflexion as healing progresses. Early functional rehabilitation

  Surgical→ Repairing the ruptured ends of the tendon suing sutures

- Kraeutler, Matthew J. Netter’s Sports Medicine. 3rd Edition. 2023.

Case 1

A 65 y/o female with hx of a.fib, PVD, and CAD comes into the ED after 12 hrs of pain and tingling in her right foot. On PE, the R. lower leg and foot are pale and cold with absent pulses however the left side has no abnormal findings. Acute limb ischemia is diagnosed and emergency surgery is performed to restore blood flow to the leg. While in recovery, she complains of intense burning pain along the lateral right leg and ankle. PE shows a normal appearing leg.

What is the most likely diagnosis?

a) Complex regional pain syndrome I
b) Complex regional pain syndrome II
c) Acute Compartment syndrome
d) Chronic Compartment syndrome
e) DVT

Answer: c) Acute Compartment syndrome

Case 1

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Which compartment is likely affected?

a) Anterior

b) Lateral

c) Superficial

d) Deep Posterior

c) Lateral

Case 1

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Which action will lead to increased pain?

a) Plantarflexion
b) Dorsiflexion
c) Inversion & dorsiflexion
d) Eversion & dorsiflexion

Answer: C) Inversion & dorsiflexion

Case 1

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Which nerve is most likely affected?

a) Deep fibular nerve
b) Superficial fibular nerve
c) Tibial Nerve

Answer B) Superficial fibular nerve

Summary Slide

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- **Know anatomy, mechanisms, symptoms, and treatments**

- Hip
  - Anatomy - blood supply. Sports injuries + degenerative joint disease.
  - Dislocations - mechanism and clinical appearance
  - Hip fractures - general mechanisms and treatment goals
  - Femur fractures - IM nail
- Knee
  - Patella fractures, instability, and dislocations
  - MCL vs. LCL, ACL vs PCL, Meniscus injuries
  - Bursitis + Baker's cyst
- Lower Leg
  - Compartment syndrome acute vs. chronic - causes, symptoms, nerves (know table on slide 57), treatments
  - Tibial fractures and DVTs
- Foot and Ankle
  - Sprains: Inversion=ATFL, eversion=deltoid ligament, high-ankle=AITFL
  - Ottawa Ankle Rules
  - Achilles tendon rupture

- Core References:

  - Moore's Clinically Oriented Anatomy, 9e, Ch. 7
  - Miller, Mark D., Orthopaedic Sports Medicine. 5th Edition. Elsevier, Inc. 2020.
  - Cleland, Joshua A. Netter's Orthopaedic Clinical Examination. 4th Edition. 2022.
  - Madden, Christopher C. Netter's Sports Medicine. 3rd Edition. 2023.